

Setup Banner Admin Page Application in Local Machine

This page includes information about how to configure a development environment for programming Banner Administrative Pages 9.x, also called “transformed” pages. This includes cloning, importing, and configuring the Banner source code.

Install Required Software

- Java Development Toolkit (1.8 or 1.7 JDK)
- Tomcat (better apache-tomcat-7.0.47)
- Eclipse Java EE IDE for Web Developers (Oxygen or Neon version)

*Apache-maven-3.2.3

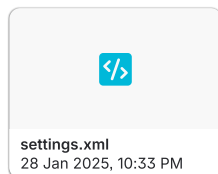
Better to create a separate directory for JDK, JRE, Apache Tomcat and Apache Maven

Maven Settings for your Development Environment

1. Create an .m2 directory under your user’s home directory. The directory may already exist so check first to see if it does.

```
mkdir c:/Users/<yourusername>/.m2
```

2. Copy the settings.xml file into this directory. I attached the file with this page.



Bring up and configure Eclipse IDE

1. Create the following directory structure for your Administrative page development.

```
mkdir c:/admin_page
cd c:/admin_page
mkdir dev90
```

2. You are going to clone the Banner Admin page GIT repository into c:/admin_page/dev90 and use the as the Eclipse workspace.

Environment Variable Settings and Path Settings

1. Create JAVA_HOME environment variable . eg : C:\extend_banner\Java\jdk1.8.0_121
2. Create JAVA_OPTS environment variable eg : -Xms1024m -Xmx2048m
3. Create JRE_HOME environment variable eg: C:\extend_banner\Java\jre1.8.0_181

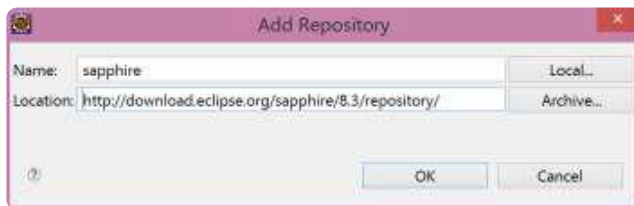
4. Create M2_HOME environment variable eg: C:\extend_banner\apache-maven-3.2.3 (give the path of maven files)
5. Create CATALINA_HOME environment variable eg: C:\extend_banner\apache-tomcat-7.0.47
6. Add this line in path variable and make sure it will be in the starting.
%JAVA_HOME%\bin;%JRE_HOME%\bin;%M2_HOME%\bin;

Install Banner Admin Page Eclipse Plugins

You must install the Foundations and Frames plugin and the Template plugin in your Eclipse installation before you can work with the Banner Administrative Pages.

1. Open Eclipse.
2. In the Workspace window, browse to or enter c:\admin_page\dev90 as your project workspace.
3. Access Help > Install New Software ... and click Add to add the plugin URL in Eclipse.
4. Install the third party Sapphire Eclipse plugin by entering a Name and the following URL in the Location for the plugin site.

<http://download.eclipse.org/sapphire/8.3/repository/>



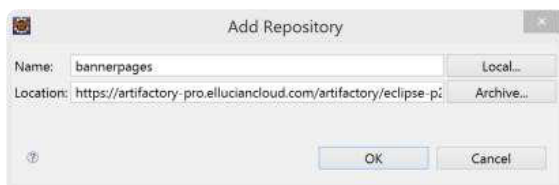
5. Click OK and select all on page to confirm the software installed.
6. Confirm license agreement.

Note: Do not restart eclipse until the next step to install the Foundations, Frames and Foundation Templates for Ellucian is finished.

7. Install the Foundations, Frames and Foundation Templates for Ellucian by entering a local Name and the following URL in the Location (URL) for the plugin site. You can choose any name, but the location must be one of the following URLs:

- Enter the URL:

<https://artifactory-pro.elluciancloud.com/artifactory/eclipse-p2-local/bannerpages-mars/>



8. Click OK.
9. If prompted for enter your password enter the following

Artifactory Username: ellucian-internal

Artifactory Password: w3_p1ck3d_!t

In the Available Software window, click Select All to install all available plugins.

10. Click Next and accept the “Licenses” agreement.

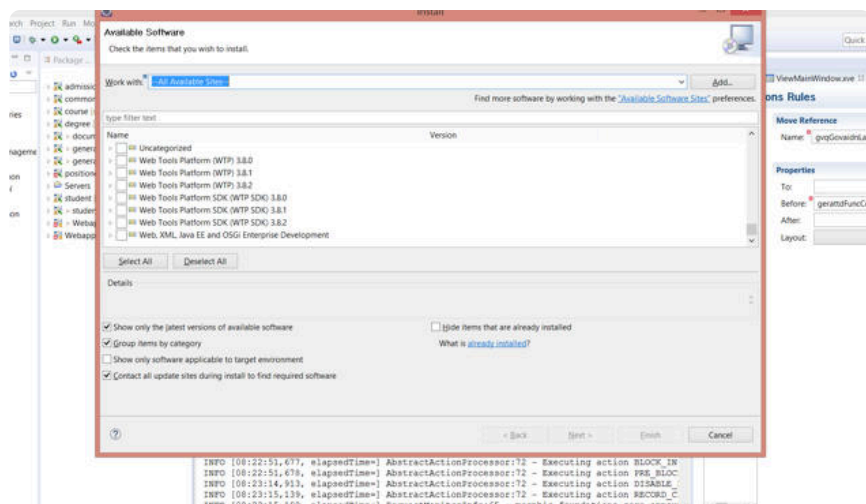
11. Click Finish.

12. If prompted with a “Security Warning” about unsigned content, click OK.

13. Again go to help menu

14. click on drop down menu arrow (left side of "Add" button)

15. Click on available sites.



16. You can see above panel. It will take a minute to load all sites.

17. Drag to last option "Web,XML,Java EE and OSGI Enterprise development". Click on the option and add the software.

18. After the install is complete, restart Eclipse. The restart is required.

Foundations Perspective

After the Foundations and Frames plugins are installed, you can view the project with the proper tools to perform Banner Admin Page development.

1. From the Window menu, select Perspective > Open Perspective > Foundations.

2. To restore the original Foundations perspective, select Perspective > Reset Perspective.

Add Server and Console Panels

Add the Server and Console panels.

1. Select Windows > Show View > Servers to show the server view.

2. Select Windows > Show View > Other > General > Console to show the console view.

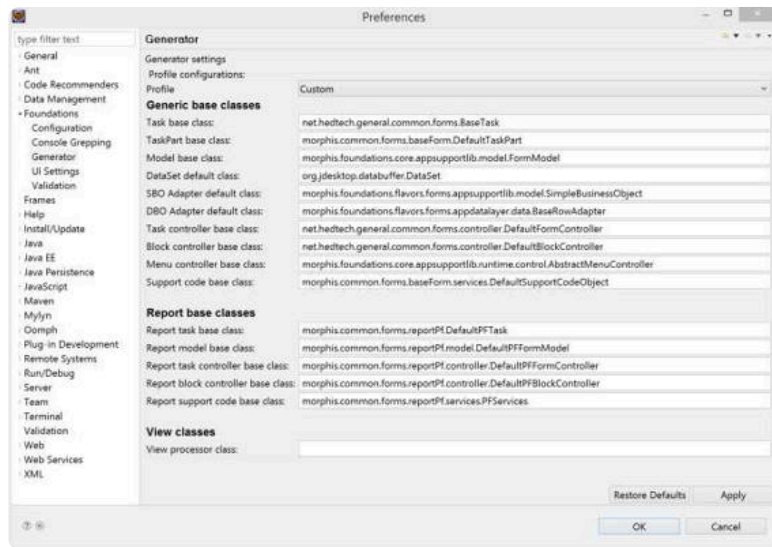
Configure Eclipse for Foundations Framework

Open the Eclipse Preferences to modify the Foundation Settings specific to Ellucian Banner Admin Pages.

1. Add Foundations Generator Settings.

1.1. Select Windows > Preferences > Foundations Generator settings.

1.2. Select the Generator tab and the Custom Profile.



Copy and paste the class names from the following table Class

Class Class name

Task base class - net.hedtech.general.common.forms.BaseTask

TaskPart base class - morphis.common.forms.baseForm.DefaultTaskPart

Model base class - morphis.foundations.core.appsupportlib.model.FormModel

DataSet default class - org.jdesktop.databuffer.DataSet

SBO Adapter default class - morphis.foundations.flavors.forms.appsupportlib.model.SimpleBusinessObject

DBO Adapter default class - morphis.foundations.flavors.forms.appdatalayer.data.BaseRowAdapter

Task controller base class - net.hedtech.general.common.forms.controller.DefaultFormController

Block controller base class - net.hedtech.general.common.forms.controller.DefaultBlockController

Menu controller base class - morphis.foundations.core.appsupportlib.runtime.control.AbstractMenuController

Support code base class - morphis.common.forms.baseForm.services.DefaultSupportCodeObject

Report task base class - morphis.common.forms.reportPf.DefaultPFTask

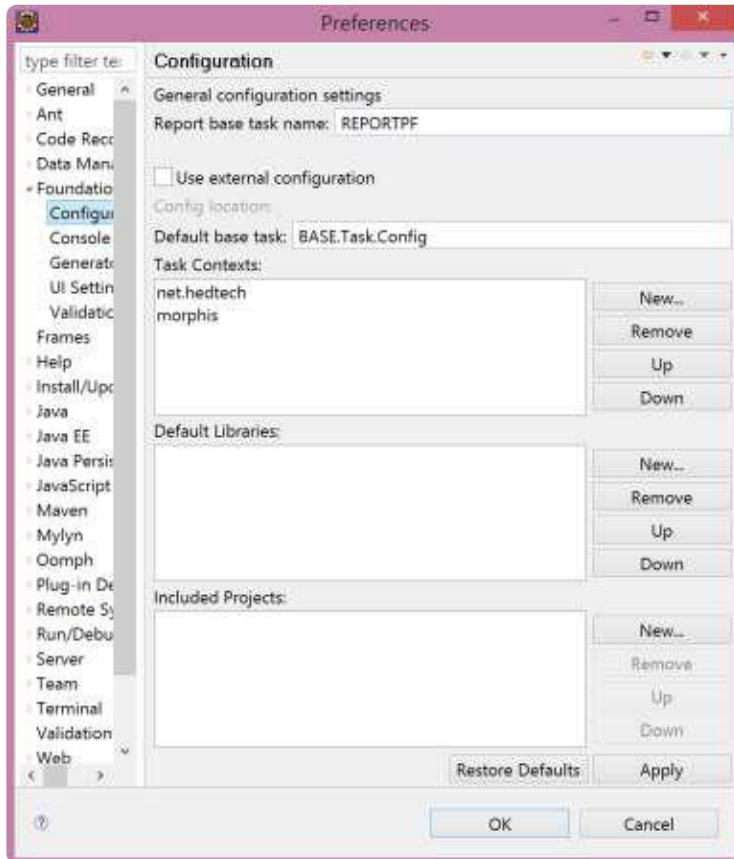
Report model base class - morphis.common.forms.reportPf.model.DefaultPFFormModel

Report task controller base class - morphis.common.forms.reportPf.controller.DefaultPFFormController

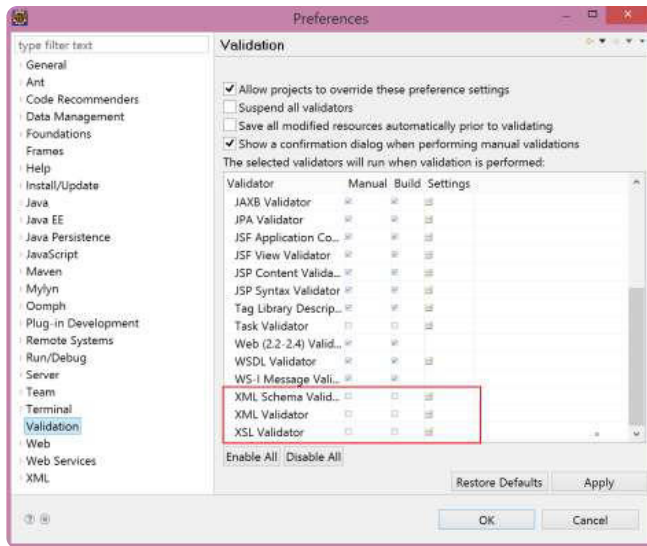
Report block controller base class - morphis.common.forms.reportPf.controller.DefaultPFBlockController

Report support code base class - morphis.common.forms.reportPf.services.PFServices

3. Select Window > Preferences > Foundations > Configuration.
4. Add net.hedtech to the list of Task Contexts.
5. Change the Default base task to BASE.Task.Config.
6. Click OK.



6. Click OK.
7. Select Window > Preferences > Validation.
8. Uncheck both the Manual and Build options for the last three Validators (XML Schema Validator, XML Validator, and XSL Validator).



9. Click Apply.

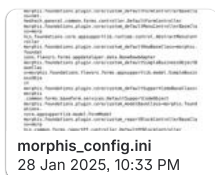
Eclipse will want to rebuild after you apply the above settings.

10. Click OK.

11. After the build finishes, exit Eclipse to prepare for cloning and importing the Ellucian repositories.

12. Eclipse allows you to import preference configurations. The following will import from a file the Foundations preferences described in this section. The file will not import Validation, Team and the Java VM parameters.

Copy the following file 'morphis_config.ini' into "eclipse\configuration" folder where your eclipse files



13. Locate the config.ini within eclipse/configuration folder and add this line to the bottom of the file.

eclipse.pluginCustomization=<path>/morphis_eclipse.ini

Clone Ellucian Banner Admin Page source repositories into Eclipse workspace

1. Clone the Ellucian GIT repositories into your C:/admin_page/dev90 directory.

```
cd c:/admin_page/dev90
```

```
git clone https://git.ellucian.com/projects/BANADMPG/repos/base/ clone dev branch
```

```
git clone https://git.ellucian.com/projects/BANADMPG/repos/build
```

```
git clone https://git.ellucian.com/projects/BANADMPG/repos/common
```

```
git clone https://git.ellucian.com/projects/BANADMPG/repos/finance
```

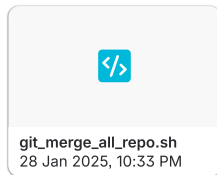
```
git clone https://git.ellucian.com/projects/BANADMPG/repos/general
```

```
git clone https://git.ellucian.com/projects/BANADMPG/repos/student
```

git clone <https://git.ellucian.com/projects/BANADMPG/repos/workspace>

or

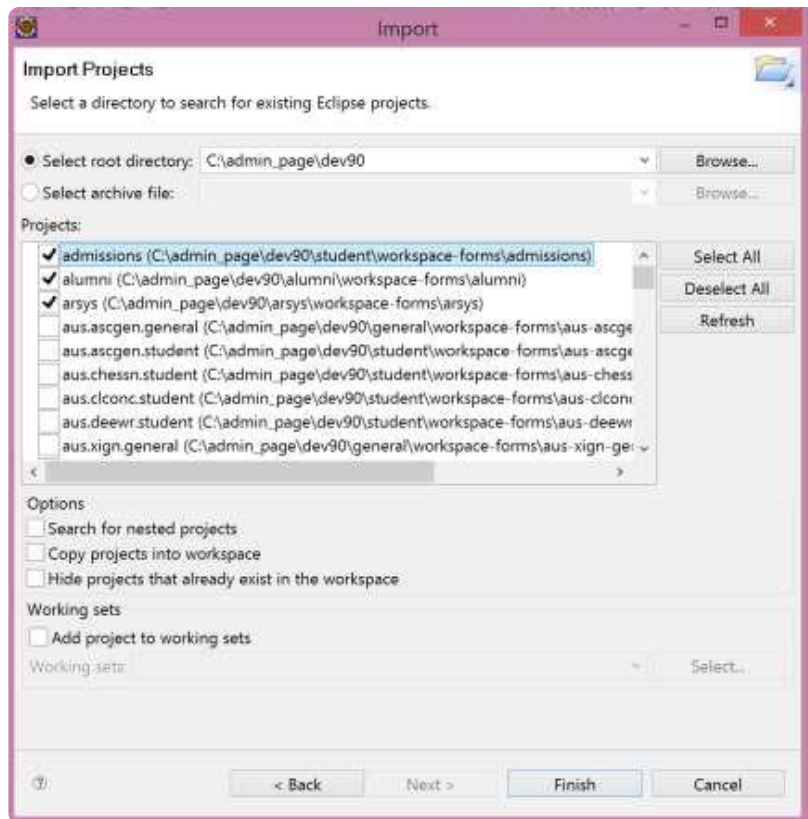
copy the following file



and double click if you have git bash.

Import Projects into Eclipse

1. Select File > Import.
2. Under General, select Existing Projects into Workspace.
3. Click Next.
4. Select the root directory of your cloned repositories c:\admin_page\dev90.



5. Click Deselect All to uncheck all projects. You may not want to import all projects because Eclipse will build everything you import and that will add extra overhead.

6. Select all the projects you want to import.

At a minimum, you need to import the following projects:

admissions

common-libraries

course

degree

document-management

student

student-common

general

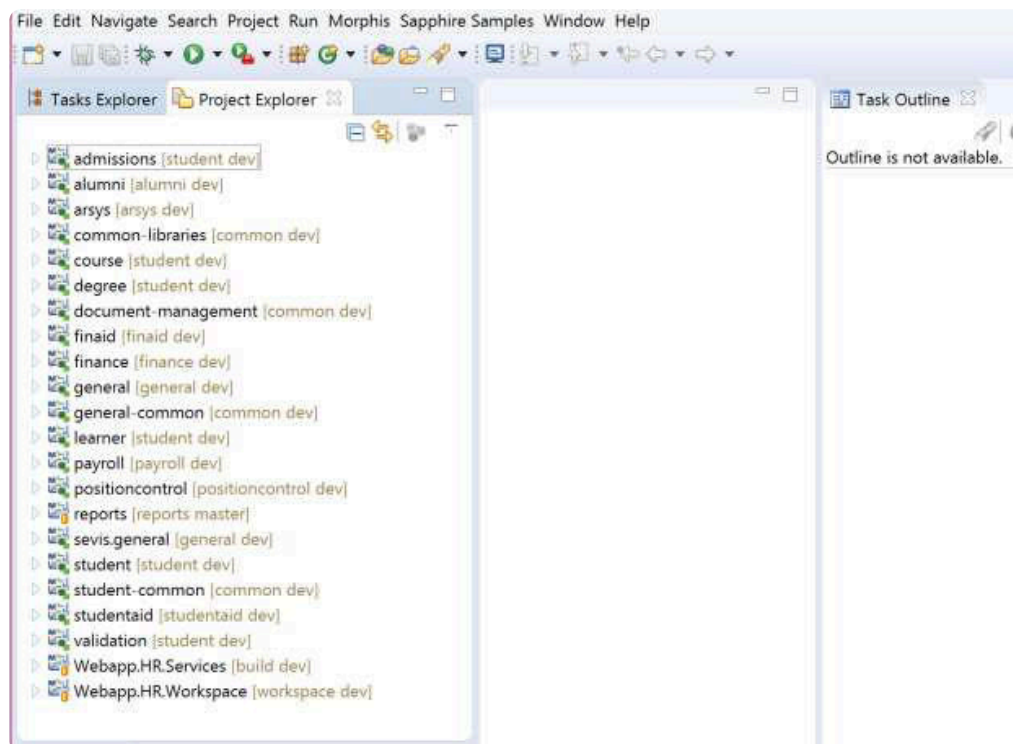
general-common

Webapp.HR.Workspace

Webapp.HR.Services (from the build repository)

7. Select Finish

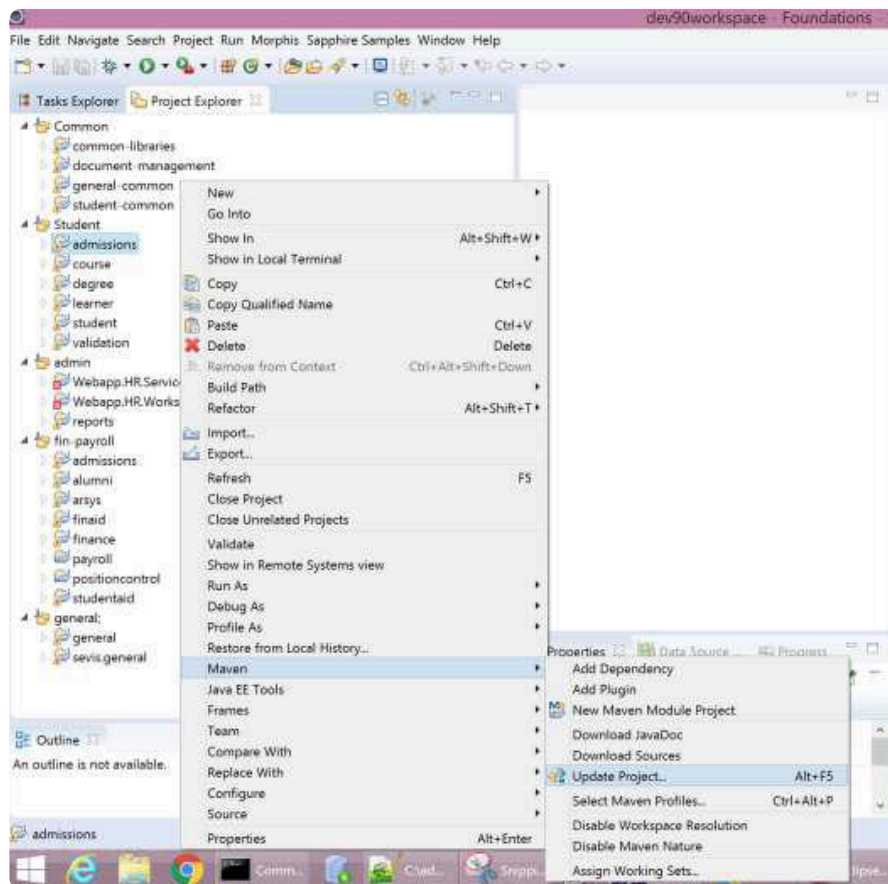
8. It will add all the projects into your workspace



Update Maven

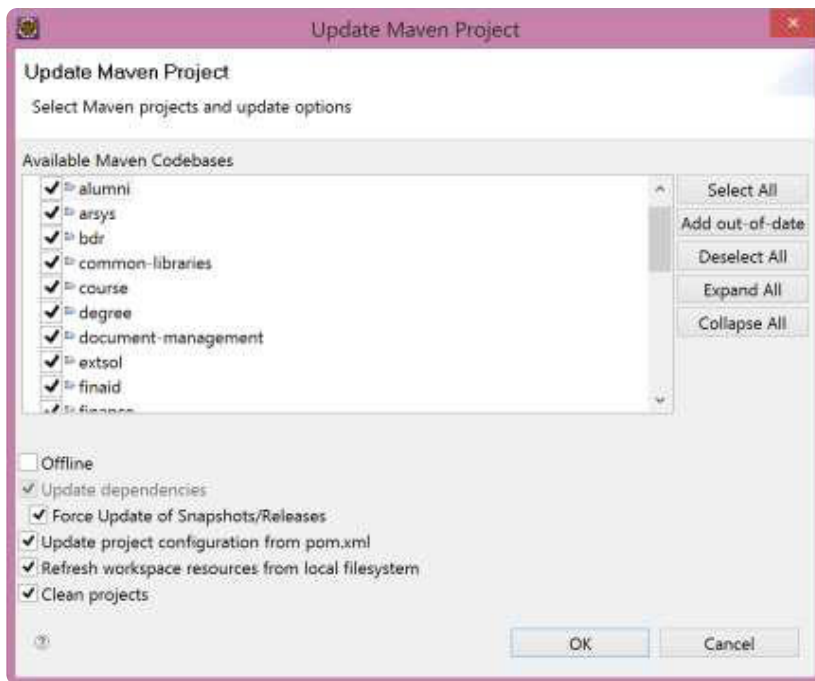
If you are setting up your environment for the first time, you need to run the Maven update process to bring in all repositories needed for Tomcat and the database server.

1. Right-click on one of your projects and select Maven > Update Project.



2. Click the Select All button to choose all projects.

3. Check the Force Update of Snapshots/Releases which will bring in current Maven repositories. The Maven repositories will be written to your .m2/repositories directory.

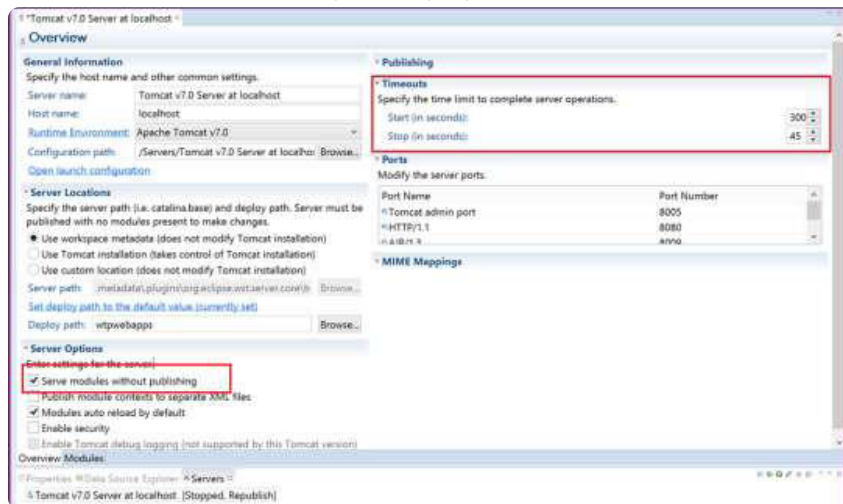


4. Click OK.

The Maven update may take a while the first time you run it.

Add Tomcat as the server for the project

1. On the Servers tab, click the No servers are available link to add Tomcat as the project server.
2. Click the Tomcat Server to open the properties and select the Overview tab.

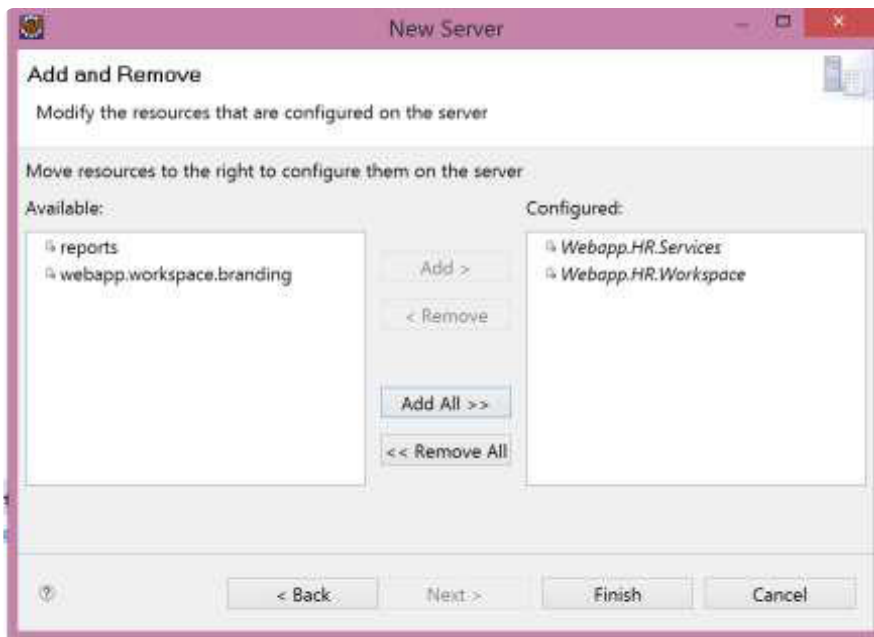


3. Make the start in seconds as 450 and stop in second 150

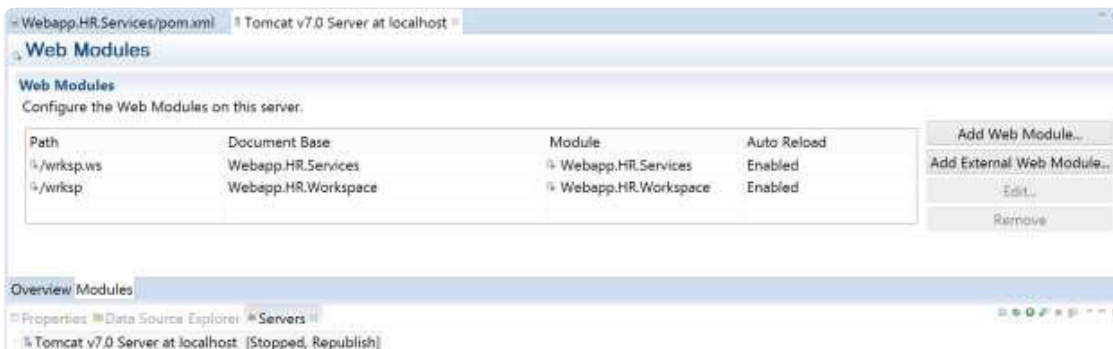
4. Expand the Timeouts and increase the Start and Stop.

5. Check the Serve modules without publishing check box.

6. Select the Modules tab and add the Webapp.HR.Services and Webapp.HR.Workspace.



7. Click Add Web Module and select the Webapp.HR.Services. If the path is not assigned, add it in [/wrksp.ws](#).
8. Click Add Web Module again, select Webapp.HR.Workspace, and assign /wrksp to the path.
9. When finished, the Modules tab will look like the following.



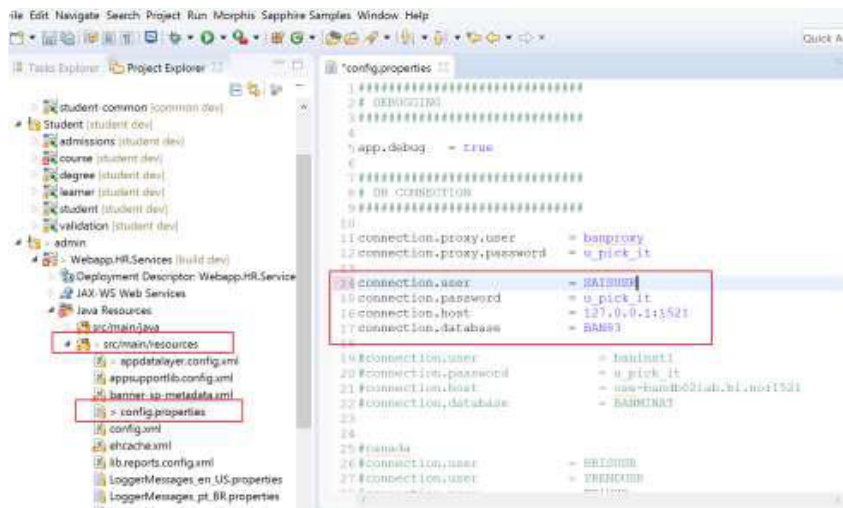
10. Save the configuration.

Ojdbc, Ucp Jar files and Tomcat

Copy the ojdbc and ucp jar files from the .m2/repositories/com/oracle/ojdbc and ucp directories to the tomcat/lib folder to resolve the two com.oracle dependencies in the pom.xml.

Project Configurations

1. Navigate to the Webapp.HR.Services project. Application specific configuration files are in the src/main/resources folder.
2. Modify the config.properties file to reflect your database information. This will be necessary for direct login to the database to succeed.



```

connection.user = HRISUSR
connection.password = u_pick_it
connection.host = 10.42.9.158:1521
connection.database = GVUDB

```

connect to this above db. Change the code accordingly.

3. Modify Appdatalayer.config.xml by uncommenting the line where you map PKG_XML to the goktxml package. This setting is usually around line 52 of the file.

```

<property name="envVars">
<map>
<entry key="PKG_XML" value="goktxml"/>
<entry key="useDBInstanceNLSSettings" value="${env.useDBInstanceNLSSettings}"/>
</map>
</property>

```

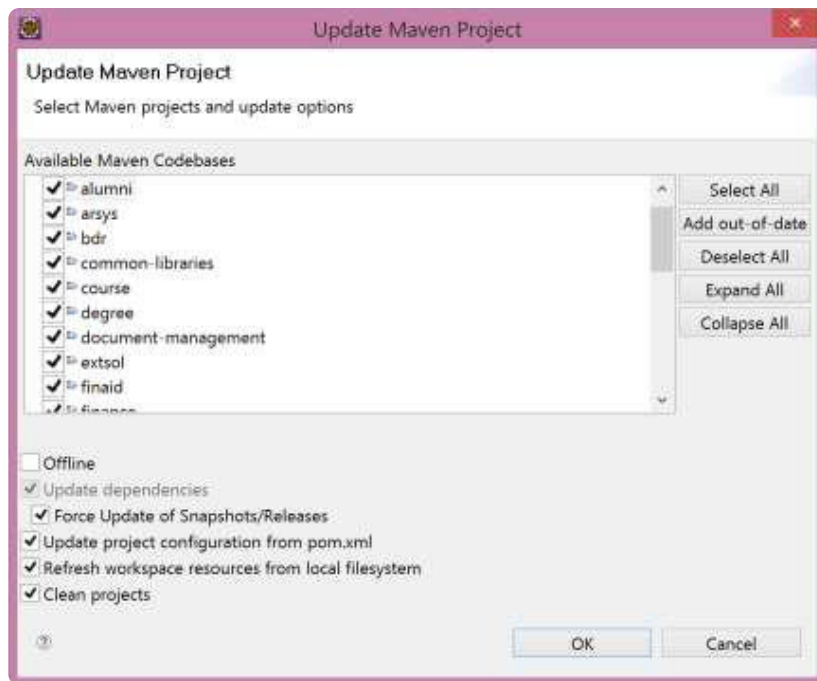
Workspace Services pom.xml

You need to set up the project dependencies in the workspace services pom.xml file before you can build the project.

Building your App

The build process compiles all pages and runs the frames framework to create the HTML pages.

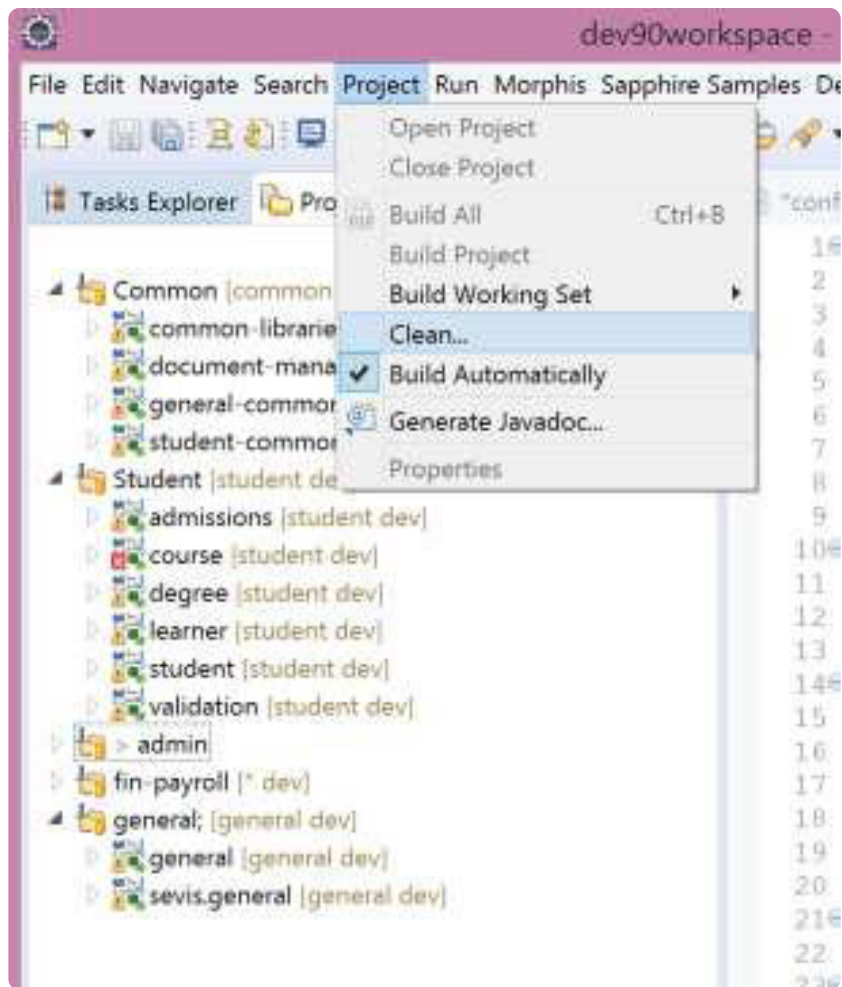
1. Right-click on one of your projects and select Maven > Update Project.
2. Click Select All to choose all projects. (If you make changes later that require a Maven update, then on subsequent runs you only need to select the projects impacted.)



3. If more updates are required, save and then perform a Project Clean. You only need to perform a project clean if you have made many changes to the source. The clean process removes cached classes and frames objects.

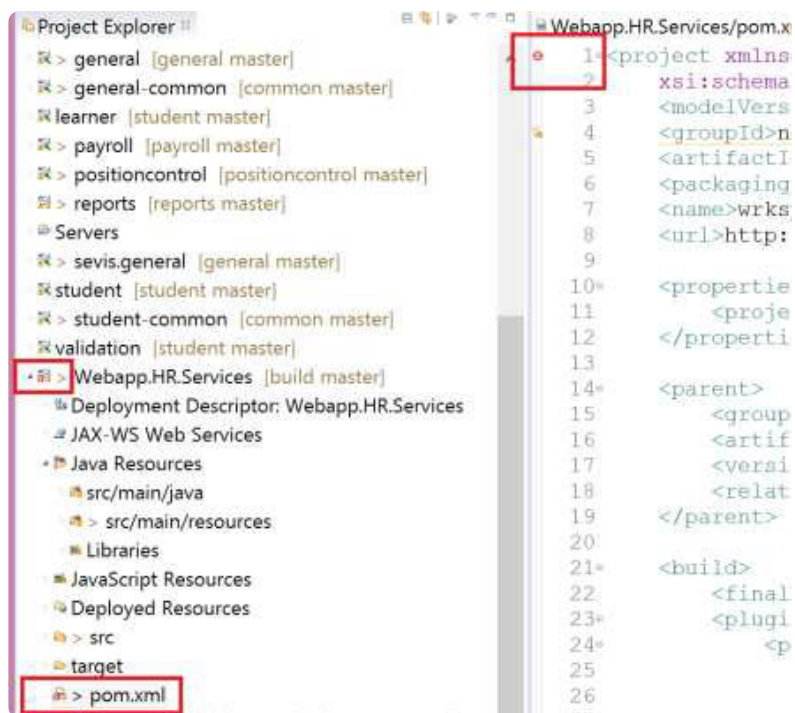
3.1. Access the Eclipse tool bar and selecting Projects > Clean.

It may take a while to complete the maven update the first time.



3.2. Select to clean all projects.

Pom dependency errors



Missing dependencies or dependencies with missing imported projects are a common problem. The build will fail and the app won't run if there is an error.

You should always check for errors with the pom files. The red x shown in the following image indicates there is an error in the pom. The error needs to be corrected before the build can continue.

Running your App

Use the following steps to start the Tomcat server from the Servers view and run the application.

1. Navigate to the following URL:

<http://localhost:8080/wrksp>

To start the app

2. Click the Tomcat server and then the start icon.

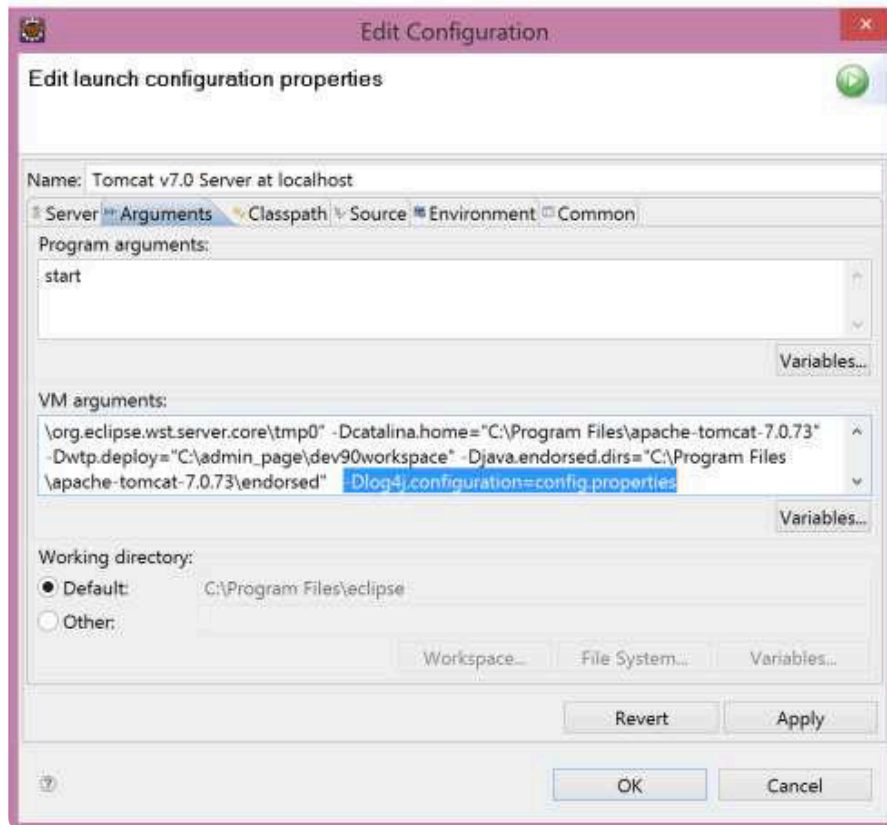
Note: Use the bug icon is to start the app in debug mode; use the arrow for a normal start.



Logging

1. From the Eclipse Tool menu, select Run > Run Configurations or click the Open Launch Configuration on the server overview properties.
2. Select the Arguments tab
3. Add the following arguments to the end of the VM arguments entry.

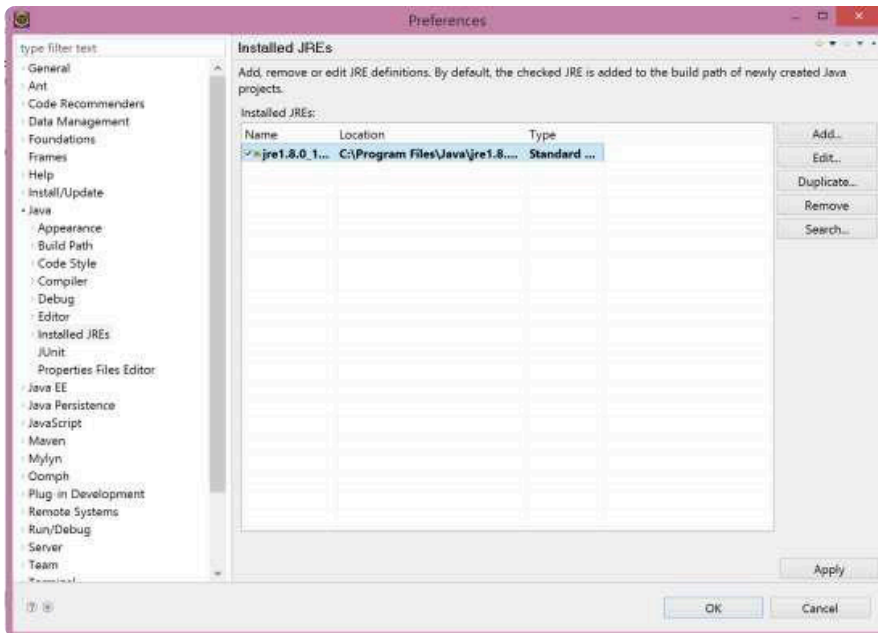
`-Dlog4j.configuration=config.properties`



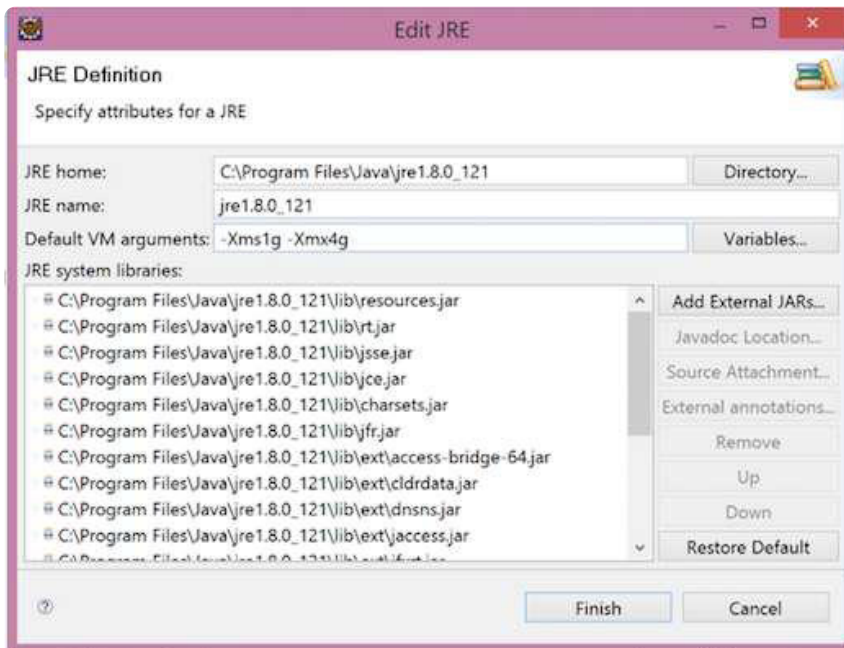
Increasing the Apps Java Heap Size

Eclipse Preferences – Update the space allocated to Java in Eclipse in the Eclipse Preferences.

- 3.1. Select Window > Preferences > Java > Installed JREs.
- 3.2. Select the JRE version you are running and click Edit.



3.3. Add the space arguments in the Default VM arguments field.



3.4. Click Finish.

